

Generation Interconnection System Impact Study Report

For Queue Project Q22-2050

Hudson Falls Hydroelectric Project 138 kV

July 29, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2050. The Interconnection Customer proposes a 106.6 MW battery energy storage facility interconnecting at the Hudson Falls Hydroelectric Project 138 kV substation in the Pennsylvania area. The request was received on April 02, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 626 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,224,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Meredosia – W Gallatin 138 kV	Felida – W Gallatin 345 kV	114.9	523	8.3
Bullmoose – Morovis 8801 138 kV	West Gallatin – Walnut 345 kV line	114.0	648	7.5
Morovis 8801 – Newcomerstown 138 kV	Walnut Substation – W Gallatin 345 kV	113.4	667	7.5
Newcomerstown – Vly Home Tap 2 138 kV line	West Gallatin – Walnut 345 kV	112.0	720	7.5
E Arlington – Vly Home Tap 2 138 kV	W Gallatin – Walnut 345 kV	111.5	741	7.5
West Gallatin 345/138kV Transformer Bank	Felida – W Gallatin 345 kV	104.9	572	8.3
Satsop – Meredosia 138 kV line	Felida Substation – West Gallatin 345 kV	104.1	458	8.6
Allegheny Hydro No 8 – Eklutna 345 kV	Piney Crk Project – Caldwell 345 kV Line	102.2	1,188	5.5
Wilson Dam – Tyrone Phelps Dodge 138 kV	Upsco – Wigwam 345 kV	101.4	461	6.2
Indian Harbour – Sheldon 138 kV	Kline Township Cogen Facility 345-138 kV Auto	101.1	242	8.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Meredosia – W Gallatin 138 kV	Rebuild line with higher-capacity conductor	\$3,288,000	12% (\$382,000)
Bullmoose – Morovis 8801 138 kV Line	Rebuild line with higher-capacity conductor	\$1,544,000	7% (\$104,000)
Morovis 8801 – Newcomerstown 138 kV	Rebuild line with higher-capacity conductor	\$4,350,000	7% (\$292,000)
Newcomerstown – Vly Home Tap 2 138 kV line	Rebuild line with higher-capacity conductor	\$2,877,000	7% (\$193,000)
Vly Home Tap 2 – E Arlington 138 kV	Rebuild line with higher-capacity conductor	\$3,066,000	7% (\$206,000)
West Gallatin 345/138kV Transformer Bank	Replace transformer with higher-rated unit	\$16,011,000	12% (\$1,859,000)
Satsop – Meredosia 138 kV line	Reconductor line	\$3,900,000	10% (\$410,000)
Allegheny Hydro No 8 – Eklutna 345 kV	Reconductor line	\$25,153,000	4% (\$948,000)
Wilson Dam – Tyrone Phelps Dodge 138 kV	Reconductor line	\$6,476,000	23% (\$1,508,000)
Sheldon – Indian Harbour 138 kV L	Reconductor line	\$1,288,000	25% (\$322,000)
	Total allocated to project		\$6,224,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.